ECON2010H

INTERMEDIATE MACROECONOMIC THEORY I

WEEK 03: CONSUMER AND FIRM BEHAVIOR

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READING: TEXTBOOK CHAPTER 4

OUTLINE

1. Introduction
2. The representative consumer
3. The representative firm

0. INTRODUCTION

- The course emphasizes the microeconomic principles.
- Two key ingredients of modern macroeconomic theory.
    - Consumer and firm behavior.
    - Aggregation.
- We start with the consumer and firm behavior, in this chapter.
- Macroeconomic theory concerns dynamic decisions, but we start with the static case.

- Consumers make choices of consumption and working.
- We want to understand how the work-leisure choice is affected by preferences and wage rates.
- Firms' decisions concern production and hiring.
- A fundamental principle is that **consumers and firms optimize**.

1. THE REPRESENTATIVE CONSUMER

- Consider a single representative consumer who acts as a stand-in for all consumers in the economy.
- Can also think that all consumers have identical preferences.
- This significantly simply models, while being adequate for many macroeconomic issues (in this course).
- Of course, consumers are diverse, as is important but beyond most topics in the course.
- We introduce the preference, budget constraint, and optimization problem of the representative consumer

1-1. PREFERENCE OF THE REPRESENTATIVE CONSUMER

- Assume that all consumers are identical in preference. So,
    - the economy behaves as if there were only one consumer.
    - we need to only focus on the behavior of one consumer.
- Assume that the consumer desires for two goods.
    - Consumption good, which we can think of as the aggregate consumption in national accounts.
    - Leisure, time spent not working in the market.

- Preference over consumption and leisure is the consumer's ranking of different bundling of the two goods

- The preference of representative consumer can be represented by a **utility function** $U(C, l)$.

    - U is the utility function.
    - C is quantity of consumption of physical goods (e.g. foods).
    - l is quantity of leisure.
- A particular combination (C, l), for example (C_1, l_1), is called a **consumption bundle**.

- Utility function gives the level of happiness or "utils", from a bundle.

- Utility function represents how the consumer ranks different consumption bundles.

- Thus, the level of utility is irrelevant, it is the relative level of utility that matters.
- Bundle (C_1, l_1) is *strictly preferred* by the consumer to bundle (C_2, l_2) if

$$U(C_1, l_1) > U(C_2, l_2).$$

- Consumer is *indifferent* between the two consumption bundles (C_1, l_1) and (C_2, l_2) if

$$U(C_1, l_1) = U(C_2, l_2).$$

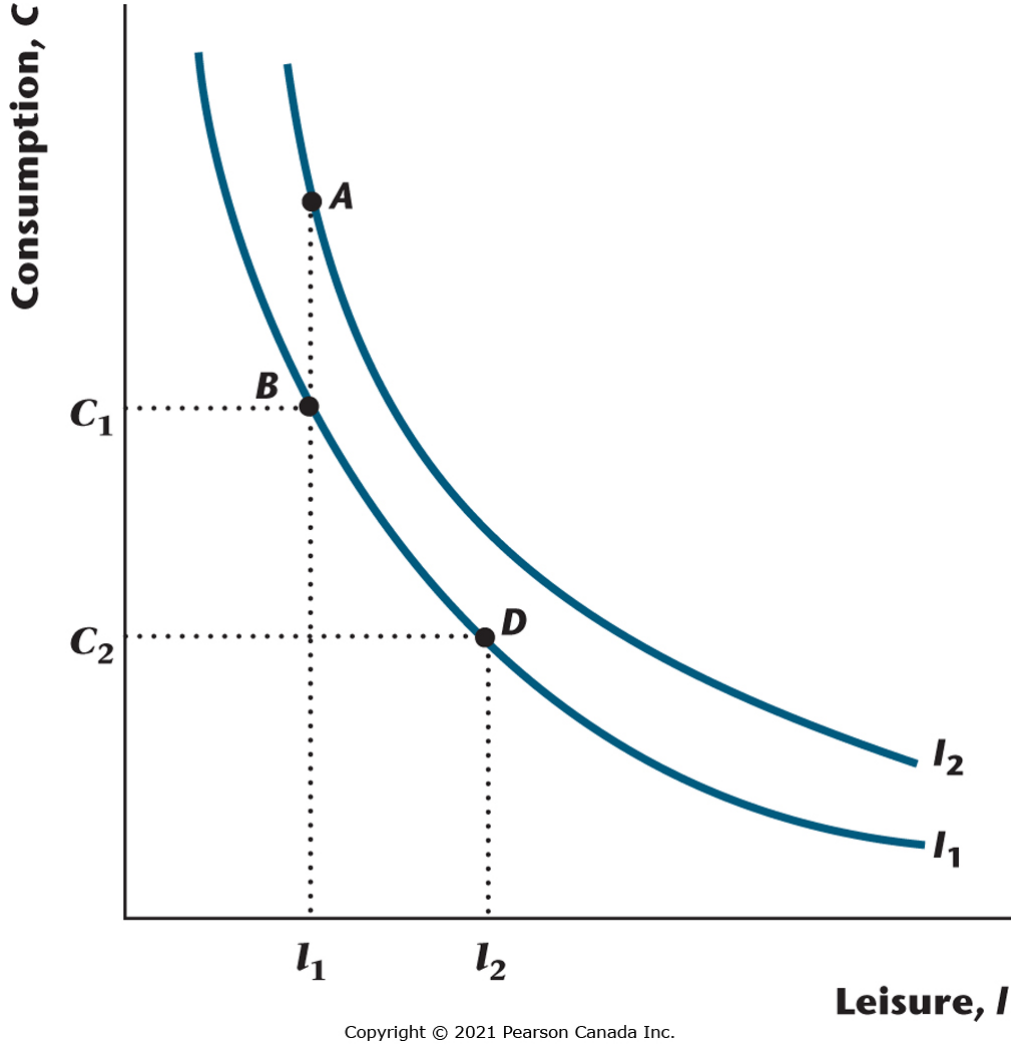
PROPERTIES OF PREFERENCES

- For a utility function to be able to represent the consumer preference, three properties of preferences should be satisfied:
    - More is preferred to less;
    - The consumer likes the diversity of the consumption bundle;
    - Consumption and leisure are normal goods.
- *More is preferred to less*: non-satiation.
    - Consumer always prefers a consumption bundle with more consumption, more leisure, or both.

- *Consumer likes diversity within a consumption bundle*.
    - Consumer prefers $(1, 2)$ to $(0, 4)$, and bundle $(1, 2)$ ranks higher than $(3, 0)$.
- *Consumption and leisure are normal goods*.
    - A good is *normal* if the amount of the good purchased by consumer increases when income increases.
    - A good is *inferior* if the amount of the good purchased *decreases* when income increases.
- Under the 3 properties of preferences, a well-behaved utility function exists to represent preference
- We study the first two properties in detail by introducing indifference curves.

INDIFFERENCE CURVE

- An **indifference curve** connects a set of points representing consumption bundles, and consumer is indifferent among these points.
- On the figure next page, I_1 and I_2 are indifference curves.
- Any point (bundle) on I_2 yields a higher level of satisfaction than any point (bundle) on I_1.
- Bundles B and D yield the same level of satisfaction, $U(C_1, l_1) = U(C_2, l_2)$.
- Bundle A yields a higher level of utility than bundle B.
    - Bundle A has the same amount of leisure as bundle B, but has a higher consumption.



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TWO PROPERTIES OF INDIFFERENCE CURVES

1. Indifference curves slope downward.

    - This follows "more is preferred to less".

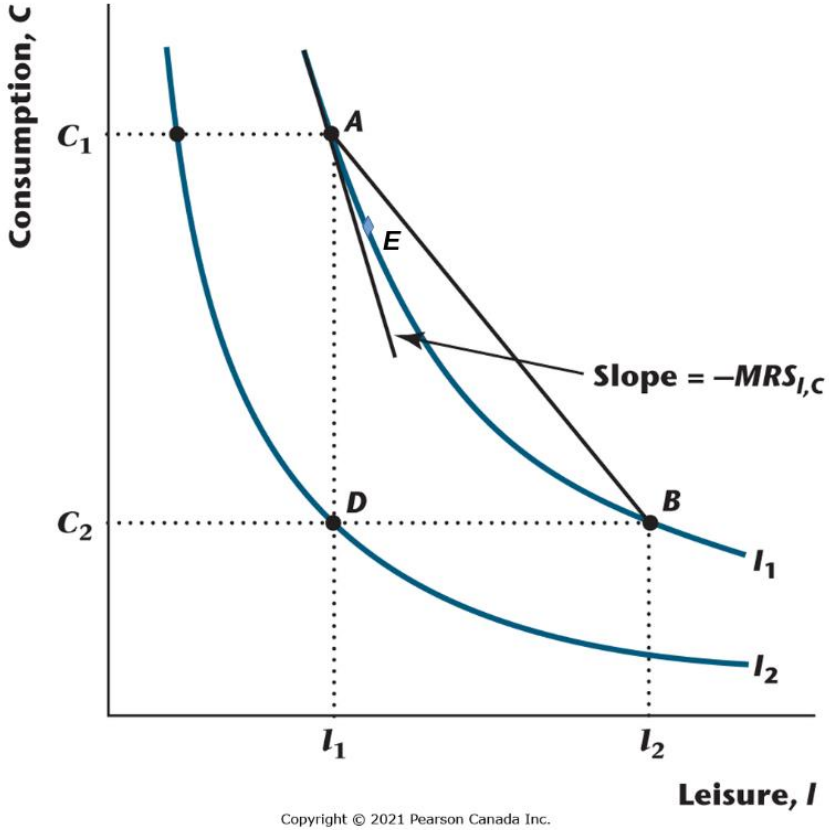
2. Indifference curves are bowed in toward the origin.

    - This follows "consumers like diversity in consumption bundles".

- The downward slopping says, if consumer needs to reduce the quantity of consumption, consumer must be given more leisure in order to keep the level of satisfaction unchanged.

    - Example, in figure on the next page, from bundle A to D then to B.

- To see why indifference curve is bowed in toward the origin, we introduce the marginal rate of substitution.



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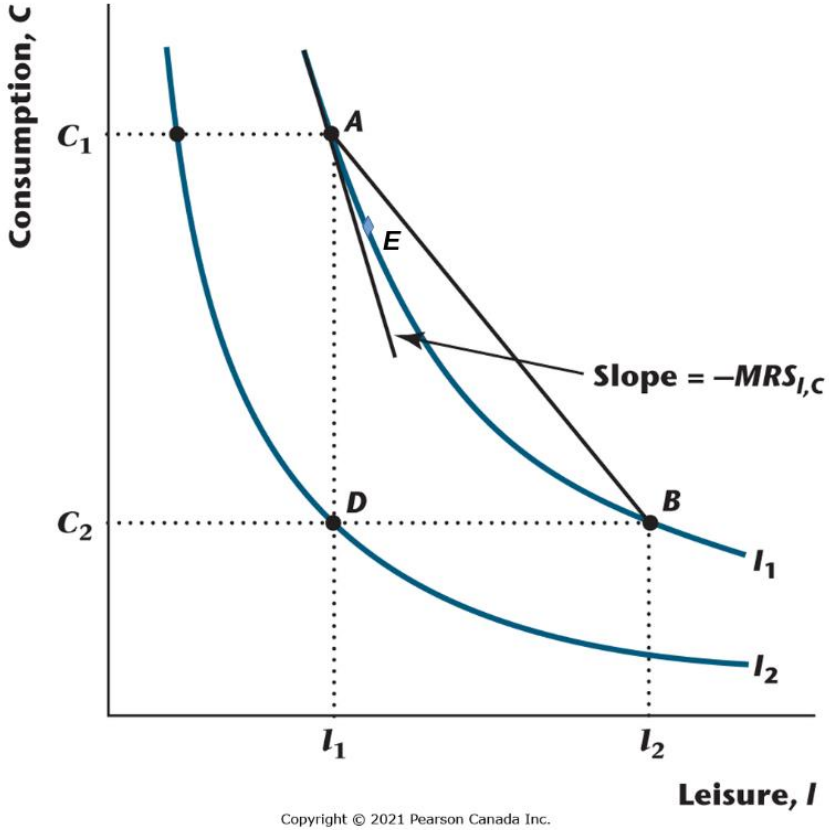
MARGINAL RATE OF SUBSTITUTION

- The **marginal rate of substitution** of leisure for consumption, denoted $MRS_{l,C}$, is the rate at which the consumer is just willing to give up consumption good per unit of increased leisure.

- Fact: $MRS_{l,C}$ **= - the slope of indifference curve passing through** (C, l).

- Why? On graph of next page, consider a point E nearby A. Let the bundle E be (C_3, l_3). The rate at which the consumer is willing to substitute leisure for consumption from A to E is the ratio $\frac{C_1 - C_3}{l_3 - l_1}$, which is $-\frac{C_3 - C_1}{l_3 - l_1}$ or the negative of slope at point A.

- **MRS is diminishing**. At point A, it takes a small amount of additional leisure if reducing consumption by one unit. At point B, it takes a larger amount of additional leisure if reducing consumption by one unit.



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- The following two statements mean the same thing:
    - Indifference curve is bowed in toward the origin.
    - Marginal rate of substitution is diminishing.
- Since the indifference curve is bowed in toward the origin, the curve becomes flatter as the bundle moves from A to B. Smaller absolute value of slope at point B means that MRS is smaller at B, relative to point A.

1-2. BUDGET CONSTRAINT OF THE REPRESENTATIVE CONSUMER

- Now a good place to make more assumptions.
- Assume that the representative consumer takes prices as given.
    - The representative consumer is a stand-in for all consumers, no monopsony (single buyer).
- Also abstract from money---the medium of exchange.
    - The means of payment for consumption and leisure is only essential when we study inflation.
- That is, we examine a **barter economy**.
    - Money price does not matter, relative price does.

- The representative consumer faces two constraints.
- The first concerns time. Consumer has h hours of time available, to be allocated to leisure l and working time N^s.
- **Time constraint**: $l + N^s = h$, also called resource constraint in general cases
- The second is the **budget constraint**,

$$C = wN^s + \pi - T.$$

- The term wN^s is real labor income, w is real wage rate.
- The term π is the real dividend income, as firms are owned by consumers.
- T is called the **lump-sum tax**, it does not depend on what the consumer has and how she/he behaves.
- This is a simplification. In reality, most taxes are not lump-sum.

- Here, "real" means that wage rate, dividend income and tax are all relative to consumption price.

- We can only add values if the above three terms the same denomination. Let P_c be the price of consumption C. The budget constraint in dollar values is

$$P_c C = W N^s + \Pi - \mathcal{T}.$$

    - W, Π, and $\mathcal{T}$ are dollar values.

- Divide both sides by P_c, we have $C = \frac{W}{P_c} N^s + \frac{\Pi}{P_c} - \frac{\mathcal{T}}{P_c}$.

- Let $w = \frac{W}{P_c}$, $\pi = \frac{\Pi}{P_c}$, and $T = \frac{\mathcal{T}}{P_c}$, then $C = wN^s + \pi - T$.

- In this economy, only w can be solved for; W and P_c separately cannot.

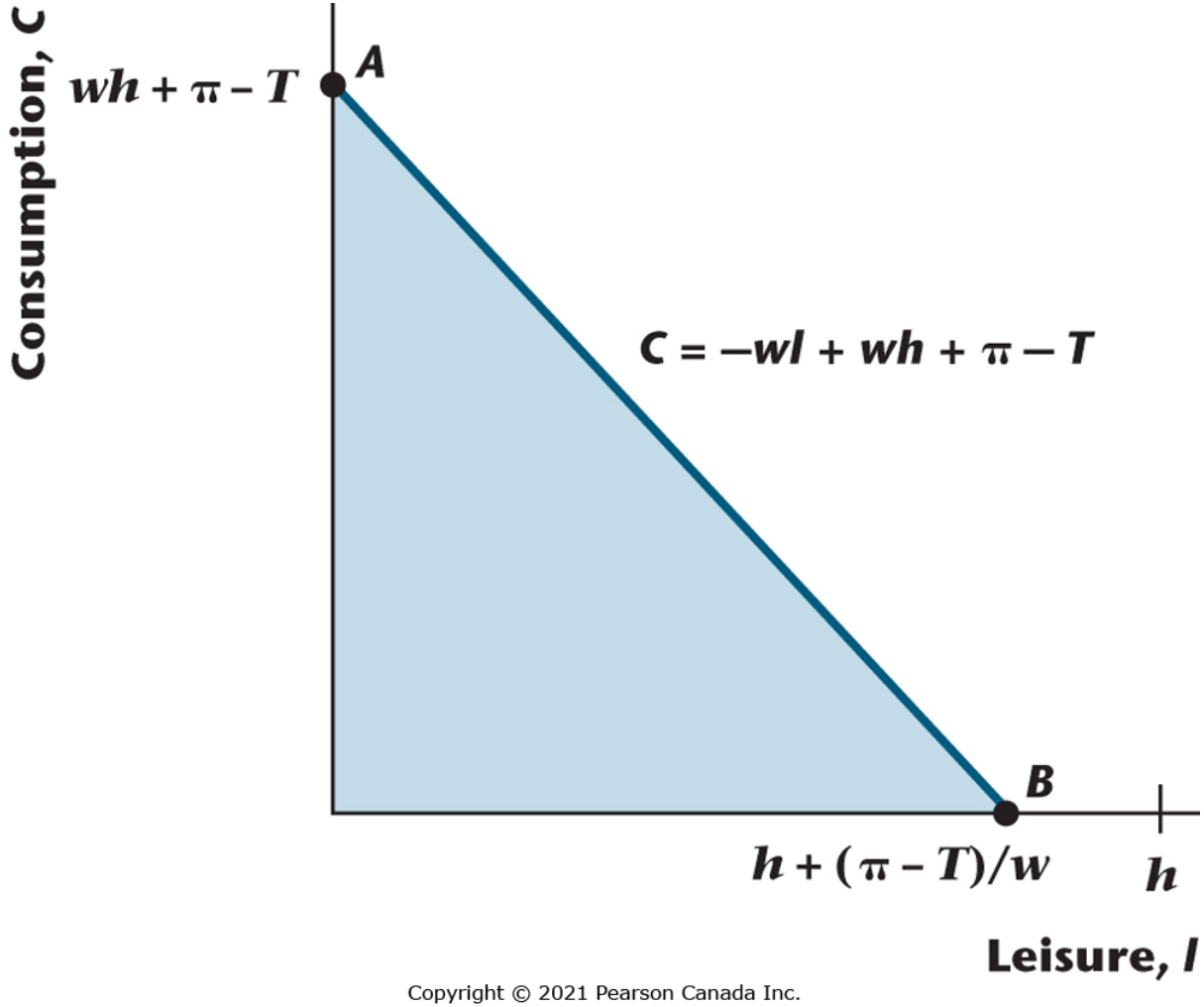
THE PRICE OF LEISURE

- Leisure is a good, it has price. Substituting $h - l$ for N^s in budget constraint, we have

$$C = w(h - l) + \pi - T.$$

- then $C + wl = wh + \pi - T$.
- On the right hand side is the total resource available at market price. The consumer can earn wh by working h hours.
- On the left hand side is the total real spending, consumption plus "spending" on leisure.
- wl is as if the consumer uses income to "purchase" leisure.
    - w is the market price of leisure time, the income of foregone hour worked.

BUDGET CONSTRAINT WHEN $T \geq \pi$



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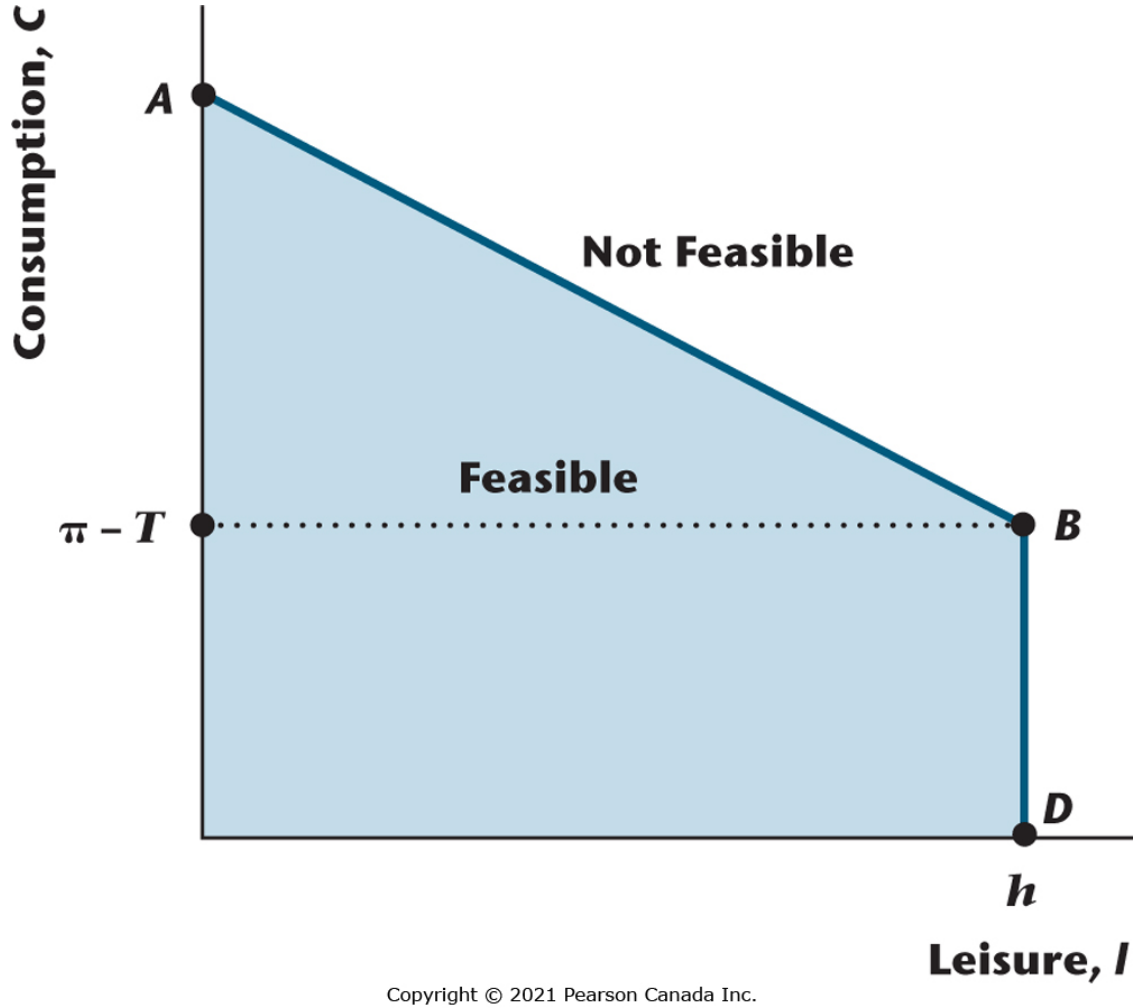
- In the above graph, the slope of the budget constraint equals $-w$.
- On the horizontal axis at point B, consumption equals zero, thus $l = h + (\pi - T)/w$.
- On the vertical axis at point A, leisure equals zero, thus, $C = wh + \pi - T$.
- The shaded area, including the straight line, is **feasible**.

BUDGET CONSTRAINT WHEN $T < \pi$



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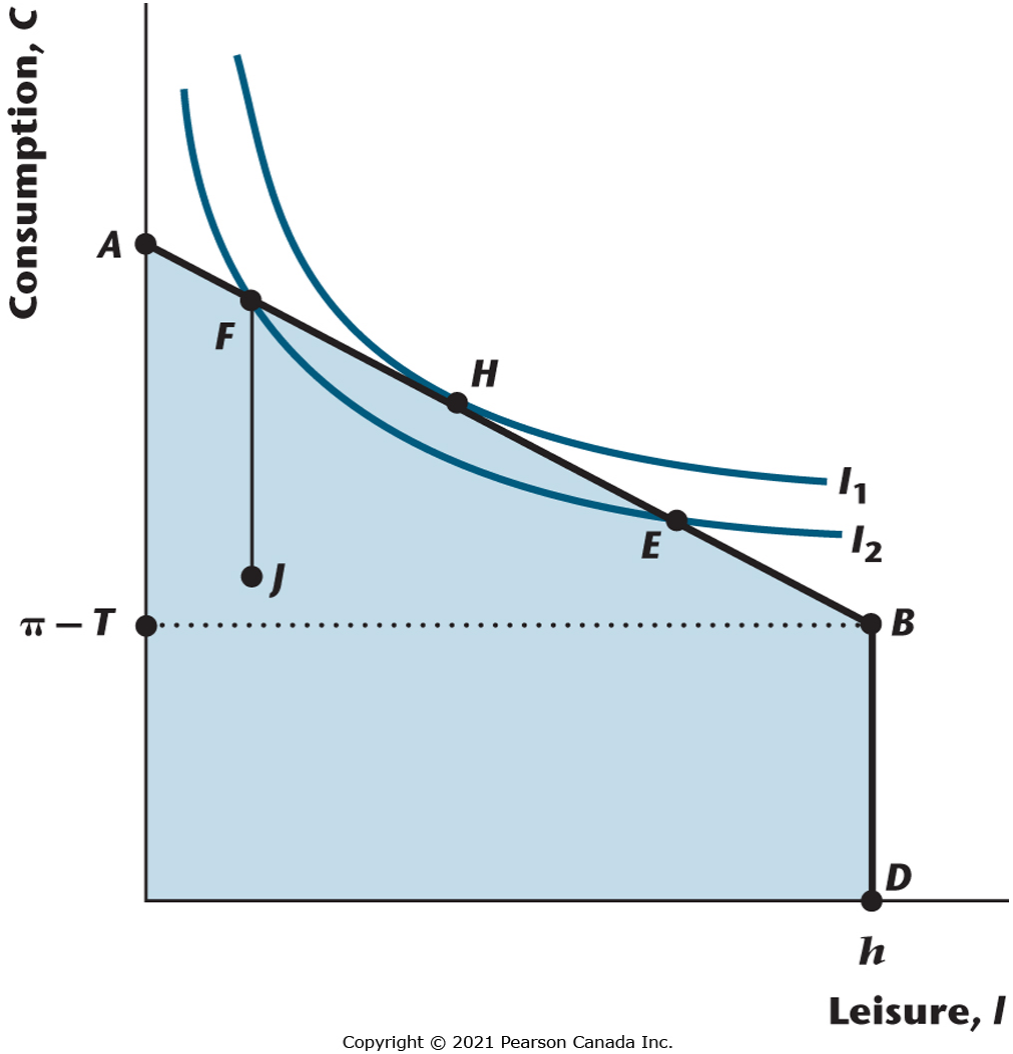
1-3. CONSUMER OPTIMIZATION

- We described the preference and budget constraint for the representative consumer.
- Now, we characterize the optimal behavior of the consumer.
- **Consumer's optimization problem is to maximize utility, given the budget constraint and resource constraint.**
- The **optimal consumption bundle** is the point (bundle) representing a consumption-leisure pair that is on the highest possible indifference curve and is on the consumer's budget constraint.
    - On a few occasions, optimal bundle can be inside budget constraint due to the nonlinearity of budget constraint
- The optimal bundle **maximizes the utility** and is **feasible**.

CONSUMER OPTIMIZATION



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- On the above graph, the farther the indifference curve moves away from the origin, the higher the utility.
- But the outward movement of indifference curve must be feasible.
- That is, the indifference curve at least has to touch the budget line.
- Therefore, point H represent the optimal bundle.
    - Any indifference curve above I_1 is infeasible.
    - Any indifference curve under I_1 is feasible but utility can be increased by moving towards I_1.

1-3-1. THE OPTIMALITY CONDITION

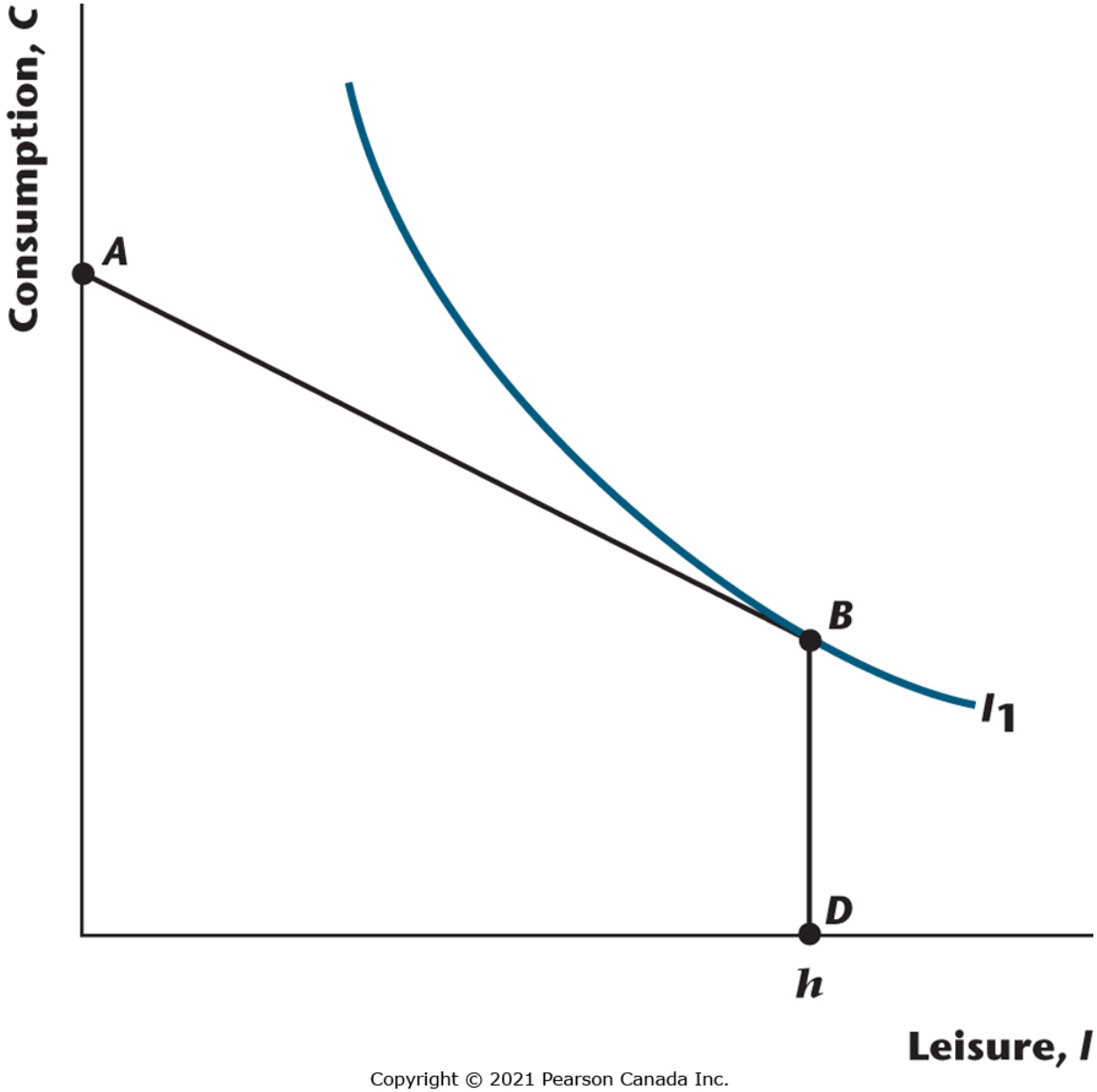
- Since at point H, the budget line is tangent to the indifference curve I_1, then

$$\mathrm{MRS}_{l,C} = w.$$

  - At the optimal bundle, the slopes of indifference curve and budget constraint (line) coincide.
- The **optimality condition**: Utility is maximized if the marginal rate of substitution of leisure l for consumption C is set to equal the relative price of l.
- When utility is maximized, then $\mathrm{MRS}_{l,C} = w$ holds.
- This optimality condition applies to a general consumer optimization problem.

WHEN DOES THE OPTIMAL l EQUAL h?

- If the optimal decision on leisure is $l^* = h$, then the optimal bundle must be at point B in the next graph.
- Choosing $l^* = h$ means that the consumer chooses not to work at all.
- However, choosing $l^* = h$ will not be possible.
    - Choosing $l^* = h$ is possible only if $\pi > T$.
    - Suppose $l^* = h$ is optimal, no output is produced since consumer does not work.
    - If no output is produced, then there is no income, and both $\pi = 0$ and $T = 0$.
    - This contradicts the condition $\pi > T$ under which $l* = h$ is feasible.



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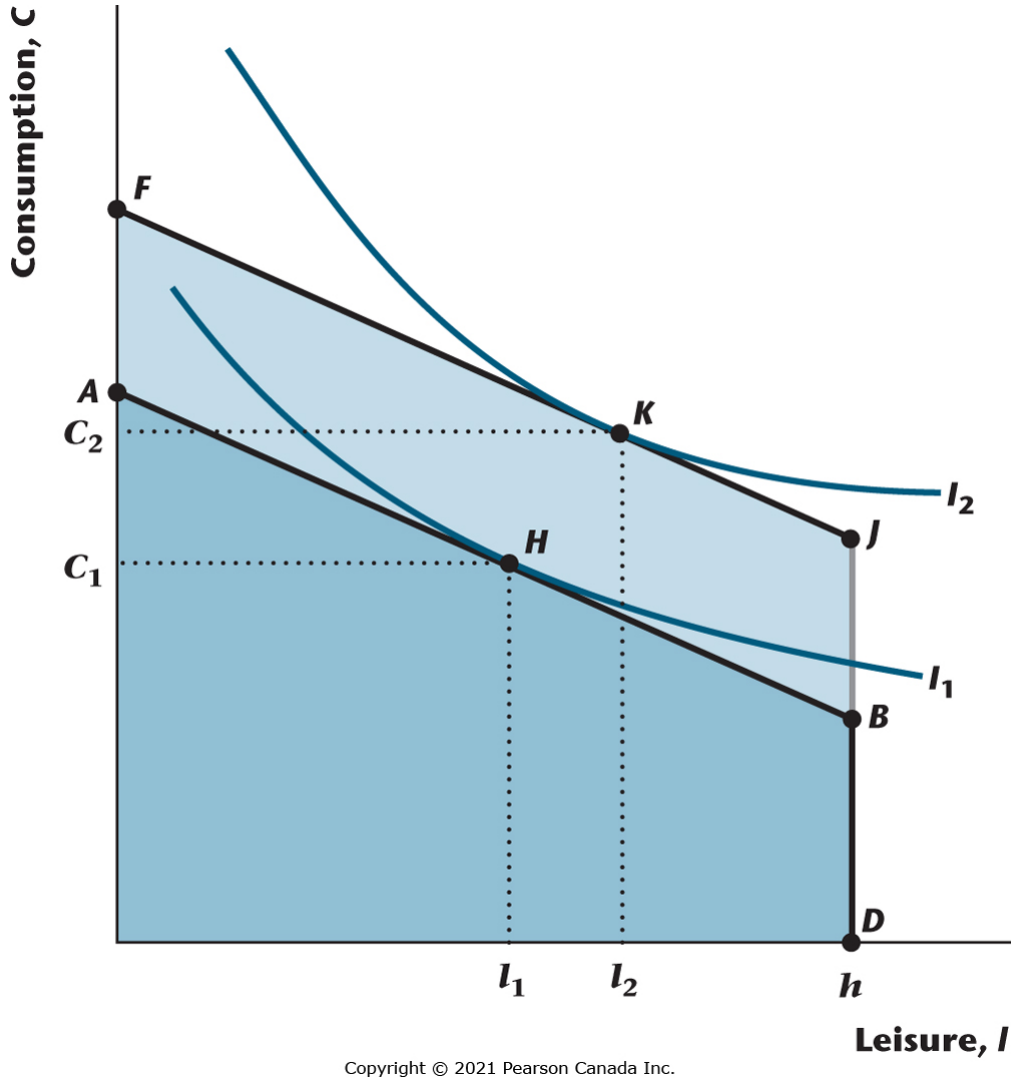
1-3-2. HOW DOES THE CONSUMER RESPOND TO CHANGES IN π OR T?

- One justification for constructing a model is to conduct "experiments",
    - for instance, what is the impact of a higher or lower tax T?
- In some sense, a model provides a laboratory apparatus.
- Consider the case of $\pi > T$. Suppose that $\pi - T$ becomes larger, arising from larger π, lower T, or both.
- The optimal bundle then moves up, both consumption and leisure increases.
- For now, we assume that real wage is not impacted by changes in $\pi - T$, which is not true in equilibrium.
- Similar conclusions for the case of $\pi < T$.



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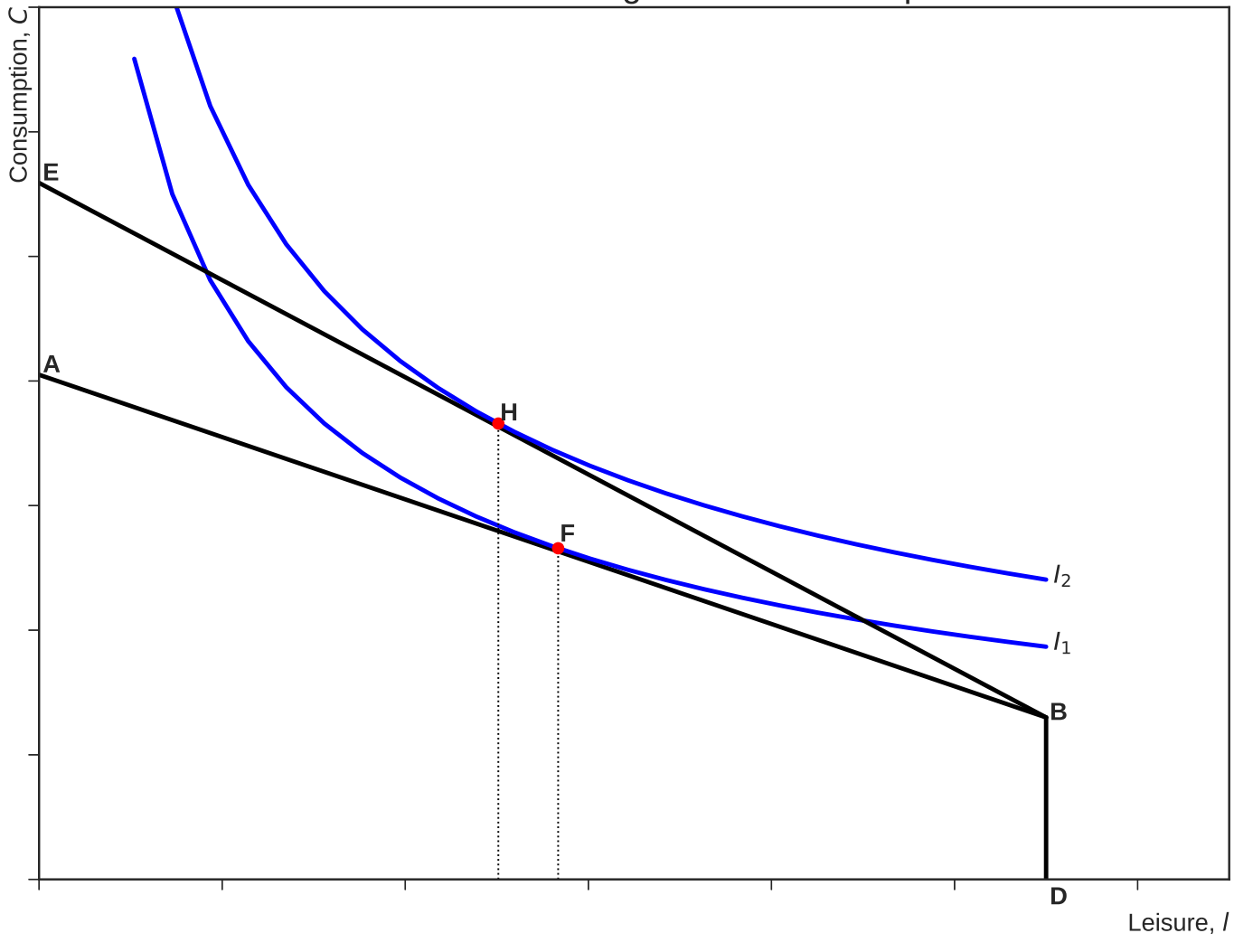
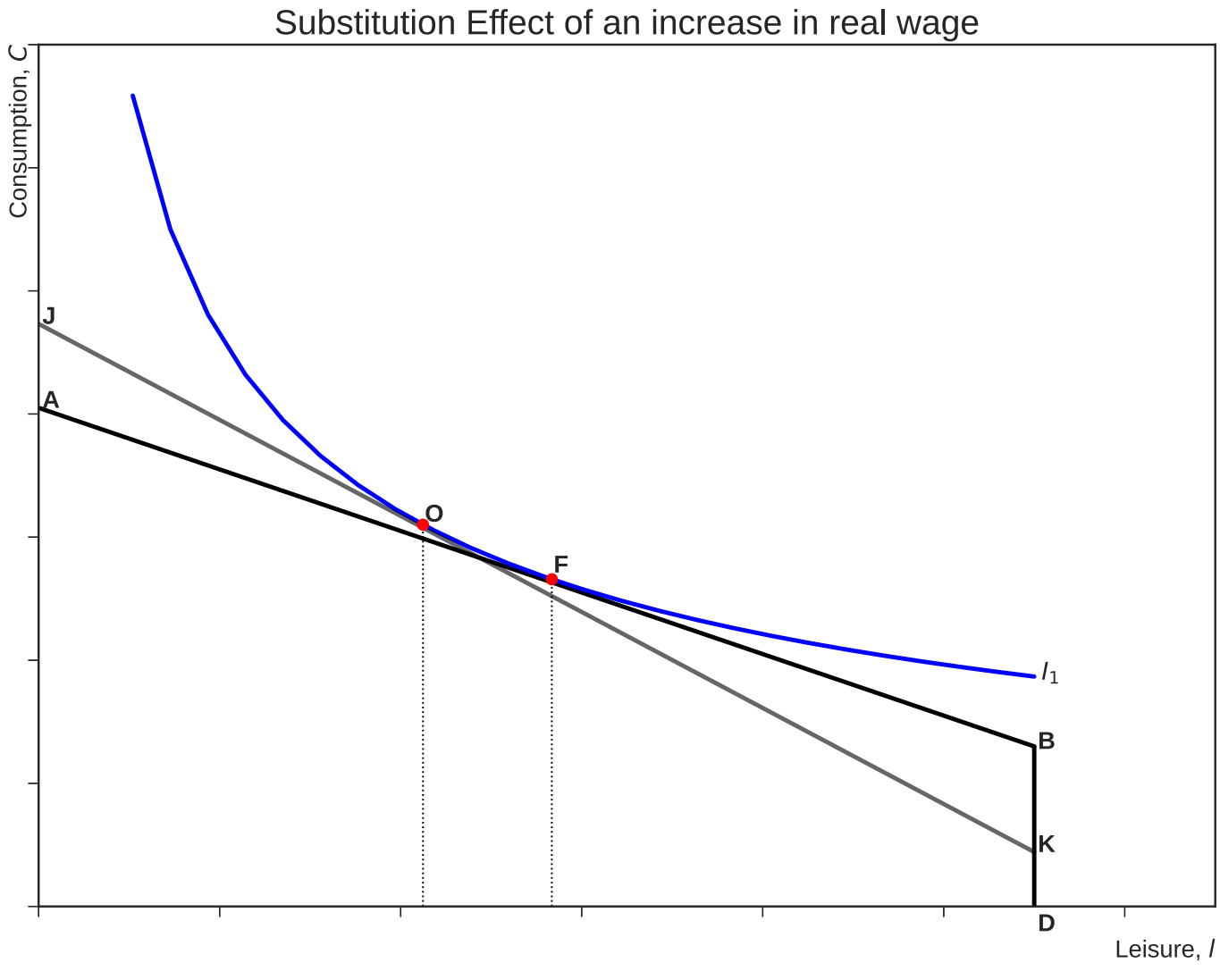
1-3-3. INCOME EFFECT AND SUBSTITUTION EFFECT

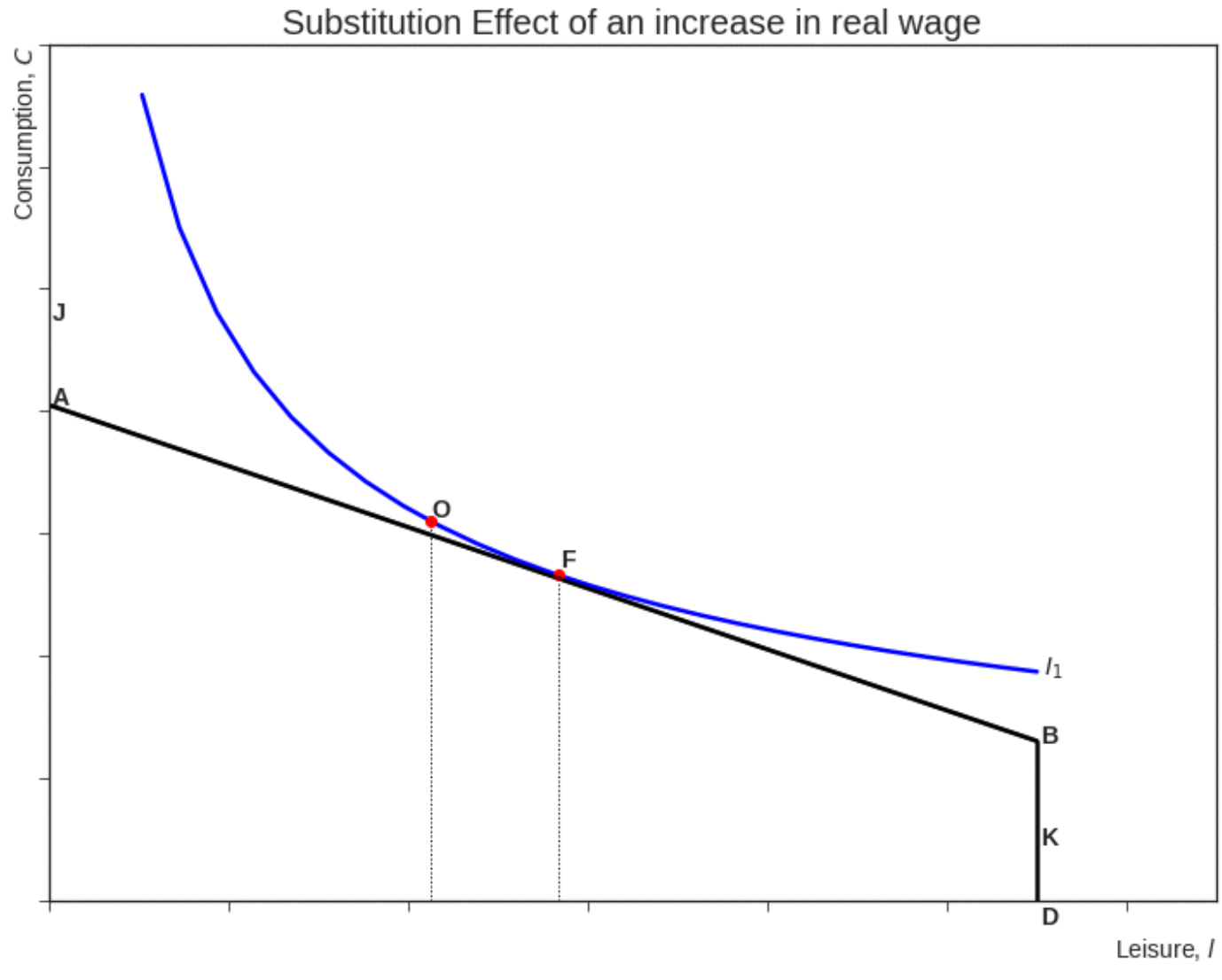
- Another experiment: **what are the effects of a change in real wage rate?**
- When real wage w rises, will consumer choose to work more or less?
- In the graph on next page, the initial optimal bundle is point F. Budget line ABD
- Point H is the optimal bundle after an increase in wage rate
- The effects of an increase in real wage on consumer's optimal choice can be broken down into
    - a substitution effect, and
    - an income effect.
- We can think the decomposition as two steps through which change in real wage affects optimal bundle.

Effect of an increase in real wage on consumer optimal choices

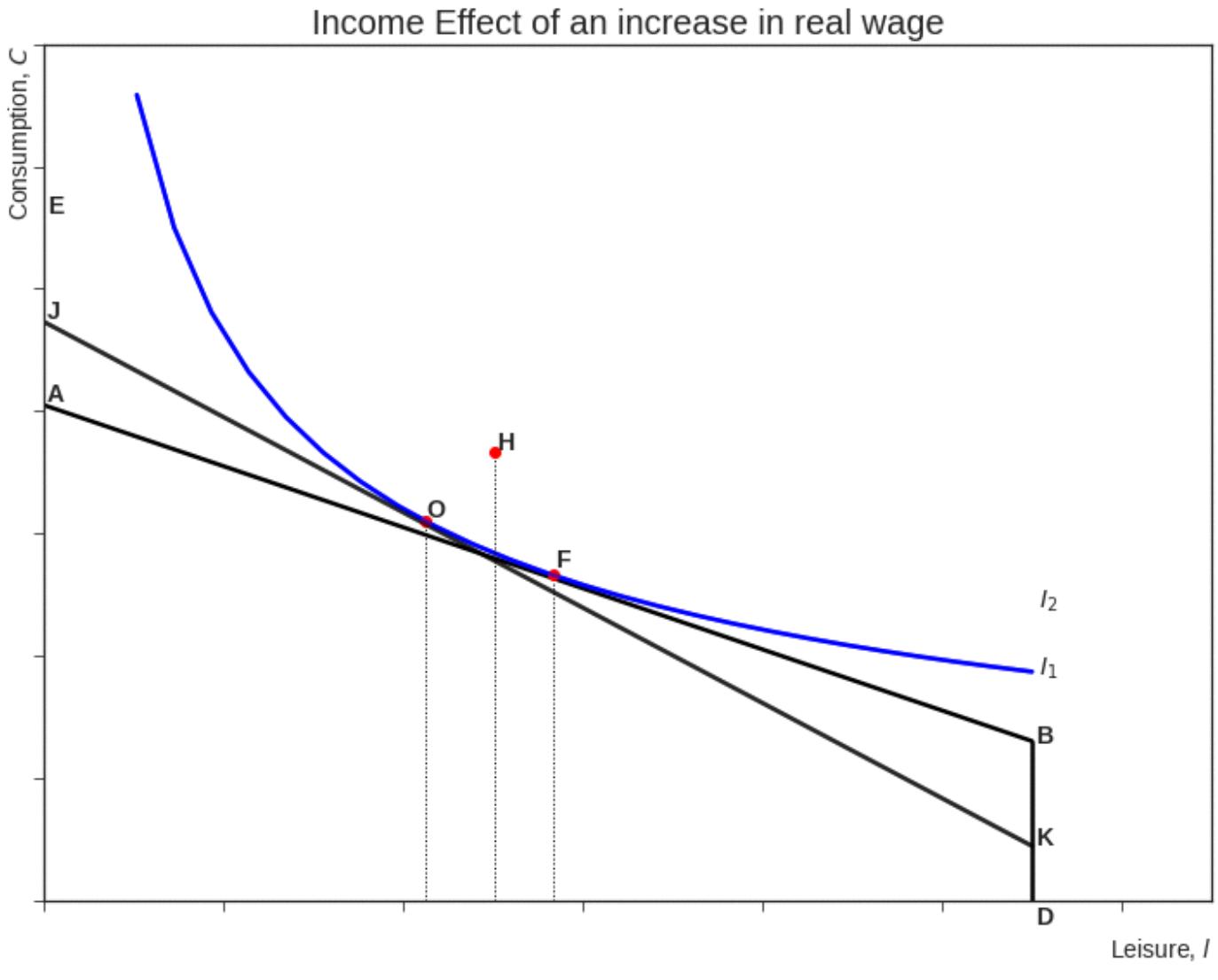


- The **substitution effect** refers to the change in the optimal bundle while maintaining the utility level unchanged, when real wage changes.
- To examine the substitution effect, suppose that the real disposable income does not change after wage increase.
    - This is a hypothetical, suppose we take away some non-wage income from consumer.
- We take away some non-wage income such that the consumer is just indifferent before and after wage increase.
    - By doing this, we only examine the change in w as *a change in the relative price of leisure*.
- After the wage increase, budget constraint becomes steeper.
- Then the optimal bundle is at point O: utility does not change from point F but B.C. becomes steeper.
- This is the substitution effect of an increase in real wage.
    - Consumption increases, leisure decreases.


Substitution Effect of an increase in real wage
Consumption, C
J
A
O
F
I_1
B
K
D
Leisure, l

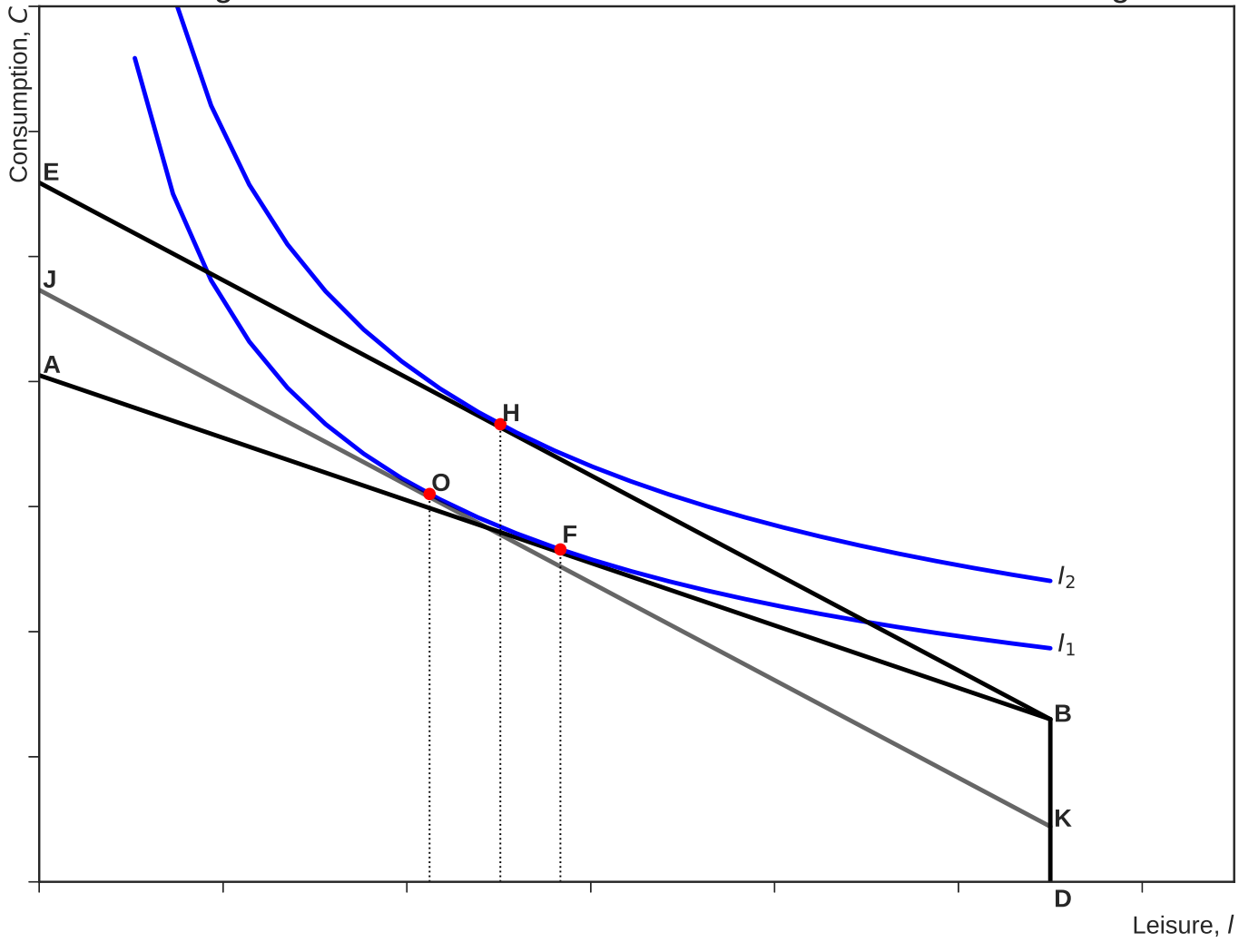

Substitution Effect of an increase in real wage
Consumption, C
J
A
O
F
I_1
B
K
D
Leisure, l

- Now we add the **income effect**,
    - the change in optimal choices by adding *income that was taken away* back to the consumer.
- From the substitution effect: the optimal choice bundle O.
- Non-wage income increases, shifting the budget line from JKD to EBD.
- The optimal choice then moves from point O to point H.
- In the income effect, both consumption and leisure increase because they are normal goods.


Income Effect of an increase in real wage
Consumption, C
E
J
A
H
O
F
I_2
I_1
B
K
D
Leisure, l

- The total effect of a wage increase equals the sum of the substitution effect and the income effect.
- On net, consumption increases when real wage rises in both effects.
- Substitution and income effects on leisure are opposite. The net effect depends on which effect is larger.
- **Following an increase in real wage, whether the consumer works more or less depends on which effect dominates**.
    - If the substitution effect dominates the income effect, the consumer will work more (less leisure).
    - if the income effect dominates the substitution effect, the consumer will work less (more leisure).
- In the graphs, substitution effect happens to dominate, consumer works more.

Adding income and substituion effects of an increase in real wage

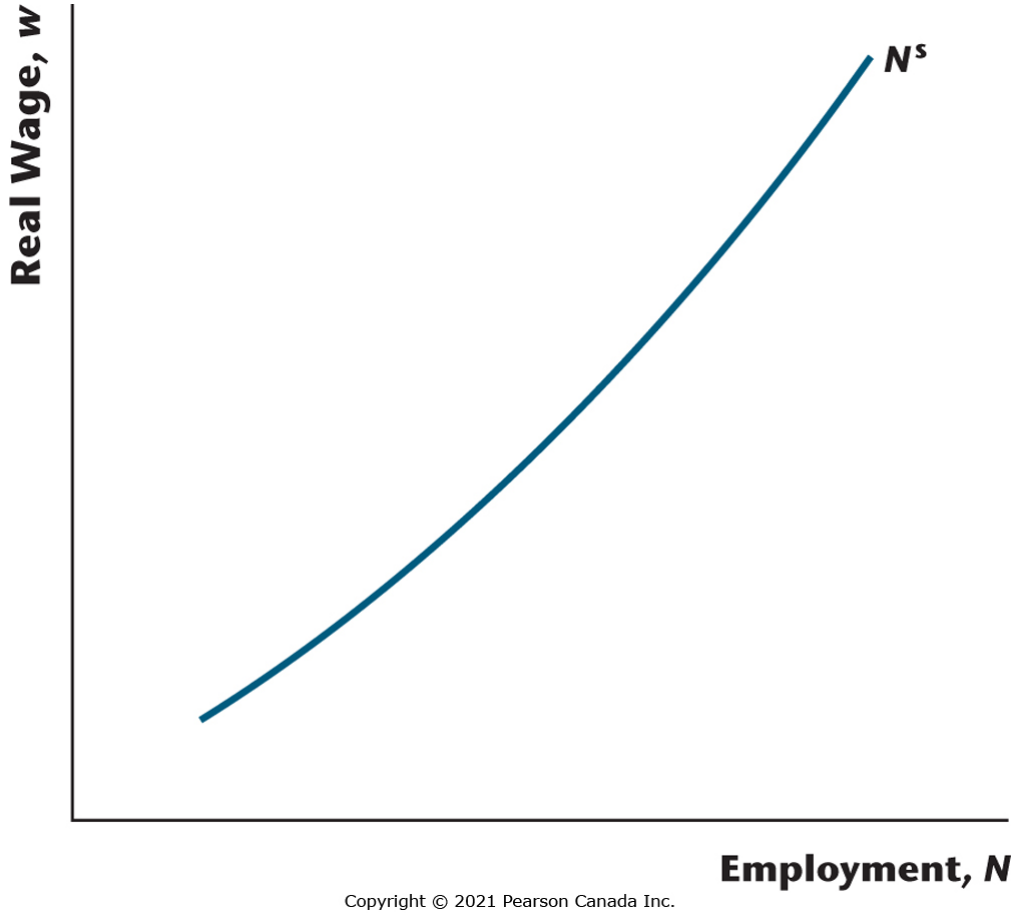


THE LABOR SUPPLY CURVE

- The **labor supply curve** describes how labor supply N^s changes as real wage w rises.

$$N^s(w) = h - l(w).$$

- The labor supply curve slopes upward if the substitution effect is larger than the income effect.
- The labor supply curve slopes downward if the income effect is larger than the substitution effect.
- Typically, in our theory, we assume that the labor supply curve has a positive slope.
- For an upward-sloping labor supply curve, an increase in $\pi - T$ shifts the curve to the left.



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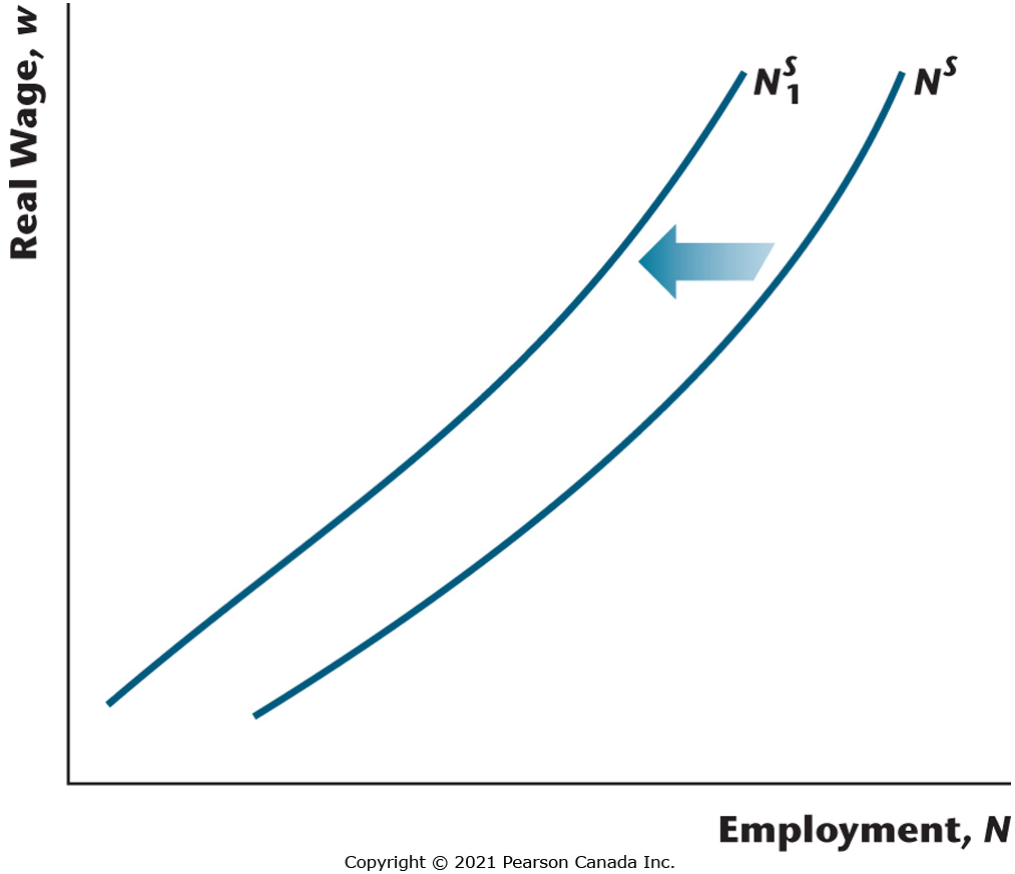

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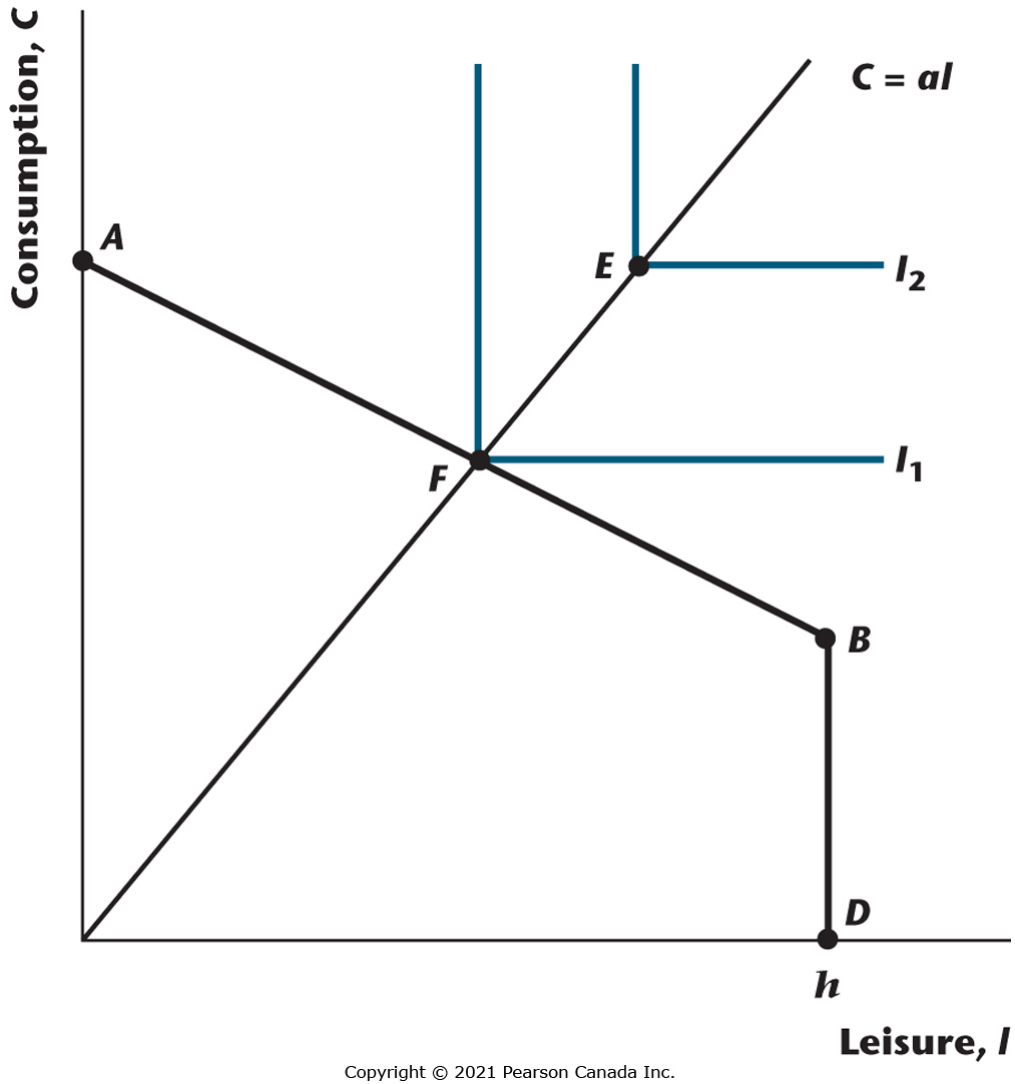
1-3-4. ANOTHER EXAMPLE: CONSUMPTION AND LEISURE ARE PERFECT COMPLEMENTS

- Consider an example of the consumer preference: perfect complements.
- Two goods are **perfect complements if the consumer always wants to consume them in fixed proportions**.
- Many examples: cellular phone and calling plan; a vehicle and gasoline (or electricity).
- Without the loss of generality, let $C/l = a$, or $C = al$ where $a > 0$ is a constant.
- With the perfect complements, the **indifference curve is L-shaped**.



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- In the graph of previous page, the optimal bundle is point F.
- The $MRS_{l,C}$ is not properly defined, as the consumer wants a fixed proporation between C and l.
- No matter how the budget line moves, the optimal bundle is always on the same line when two goods are perfect complements,
    - because $C = al$.
- Question: what is the impact if a is larger, i.e., consumer suddenly want more consumption $C?

- With the perfect complements, can solve for the optimal bundle algebraically.
- we know $C = al$, and the budget constraint is $C = w(h - l) + \pi - T$.
- These are two equations and two unkowns.
- Then $(a + w)l = wh + \pi - T$, or $l^o = \frac{wh\pi - T}{a+w}$.
- From the budget constraint, the optimal consumption is $C^o = \frac{a(wh+\pi-T)}{a+w}$.
- The optimal labor supply is $N^o = h - l^o = h - l(w)$.
- Notation: l is variable representing leisure, it takes values from 0 to h. l^o is the optimal value of l. In textbook $l(w)$ is the optimal value of l, as a function of w.

MAIN TAKEAWAY FROM THE CONSUMER'S PROBLEM

- Properties of preferences.
- Indifference curve.
- Marginal rate of substitution.
- The optimality condition.
- Substitution and income effects of a change in real wage.
- Labor supply curve.

2. THE REPRESENTATIVE FIRM

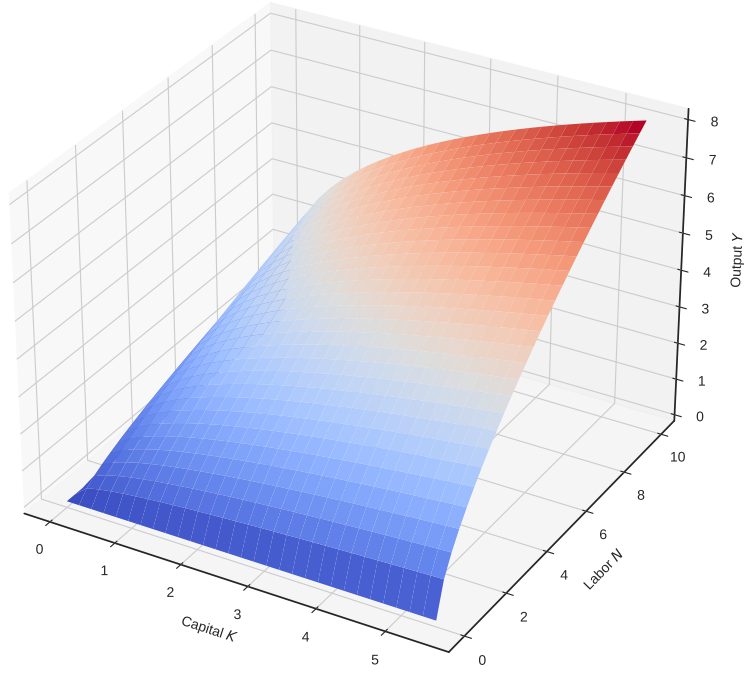
- In representative consumer's problem, we characterize the demand for consumption goods and the labor supply.
- On the supply side, in macroeconomics, real GDP is the primary measure of aggregate output.
- Real GDP equals the aggregate value added.
- Therefore, the production and the supply of final goods are value added.
- We take real GDP as one good. The production can be represented as $Y = zF(K, N^d)$.
    - Y is aggregate output.
    - z is total factor productivity (TFP).
    - K is capital stock, N^d is the quantity of labor input.
    - $F()$ is a function with two arguments.

- Intermediate inputs are not needed since the output measure is value added.
- Two issues to address:
    - What is the optimal supply, *given the factor prices*?
    - How many firms are needed to produce the aggregate output?
- For the second issue, we assume that the aggregate output is produced by a single representative firm.
- With one typical function form of F, the representative firm is well justified (more later).
- Markets are perfectly competitive.

2-1. PROPERTIES OF PRODUCTION FUNCTION

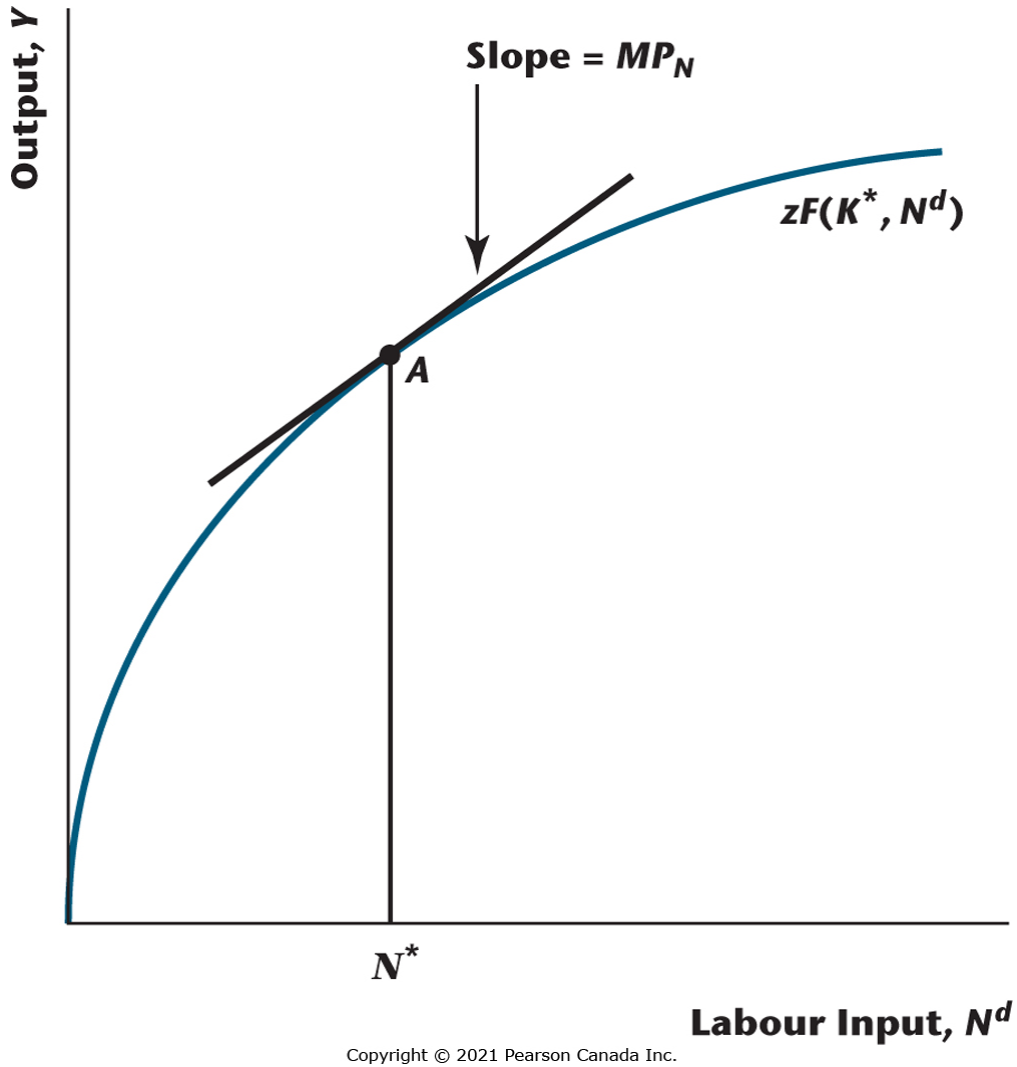
- Now come back to production function $Y = zF(K, N^d)$.
- K and N^d are called the factors of production.
- Output Y increases when K or N^d increases, or both increase.
- Adjusting K (e.g., intalling machines) may take time, thus K may be fixed in the short-run.
- Adjusting N^d (i.e., hiring and firing) appears feasible in any time horizon.
- TFP z is a shifting factor, Y increases proportionally with z.
- For now, z is given. In the long run, the economy can invest to raise z.

Cobb-Douglas production function: $Y = zK^{\alpha}(N^{d})^{1-\alpha}$ with $0 < \alpha < 1$



MARGINAL PRODUCT

- The **marginal product** of a factor of production is the additional units of output that can be produced with one additional unit of that factor, holding constant the quantities of the other factors of production.
- Marginal product of labor input, MP_N, is the slope of production function when fixing capital at a constant $\overline{K}$.
- Marginal product of capital input, MP_K, is also a slope.
- **Marginal product of a factor declines as more units of that factor are used**.
- This is a result of assumptions on the production function, coming next.



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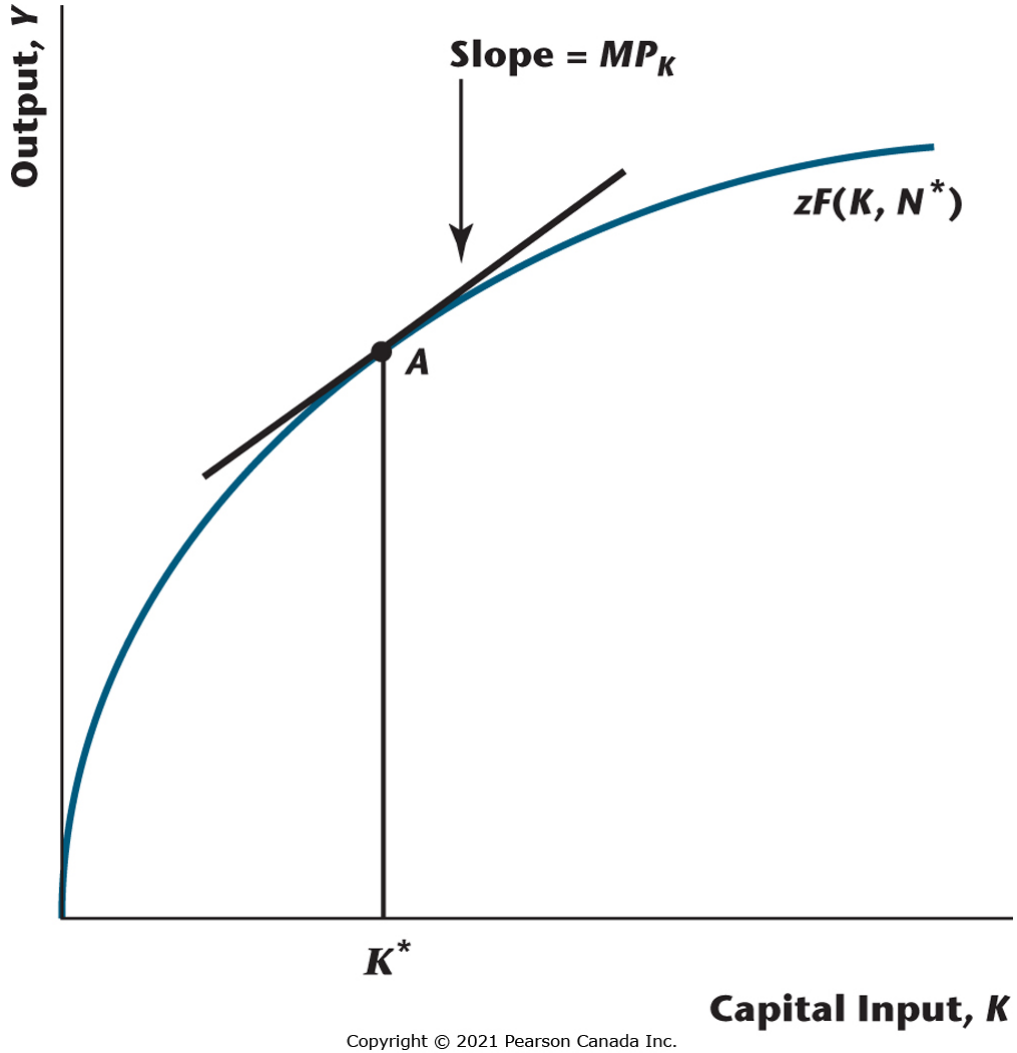

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PROPERTIES OF PRODUCTION FUNCTION

- **Property 1.** The production function exhibits **constant returns to scale (CRS)**: for any $x > 0$,

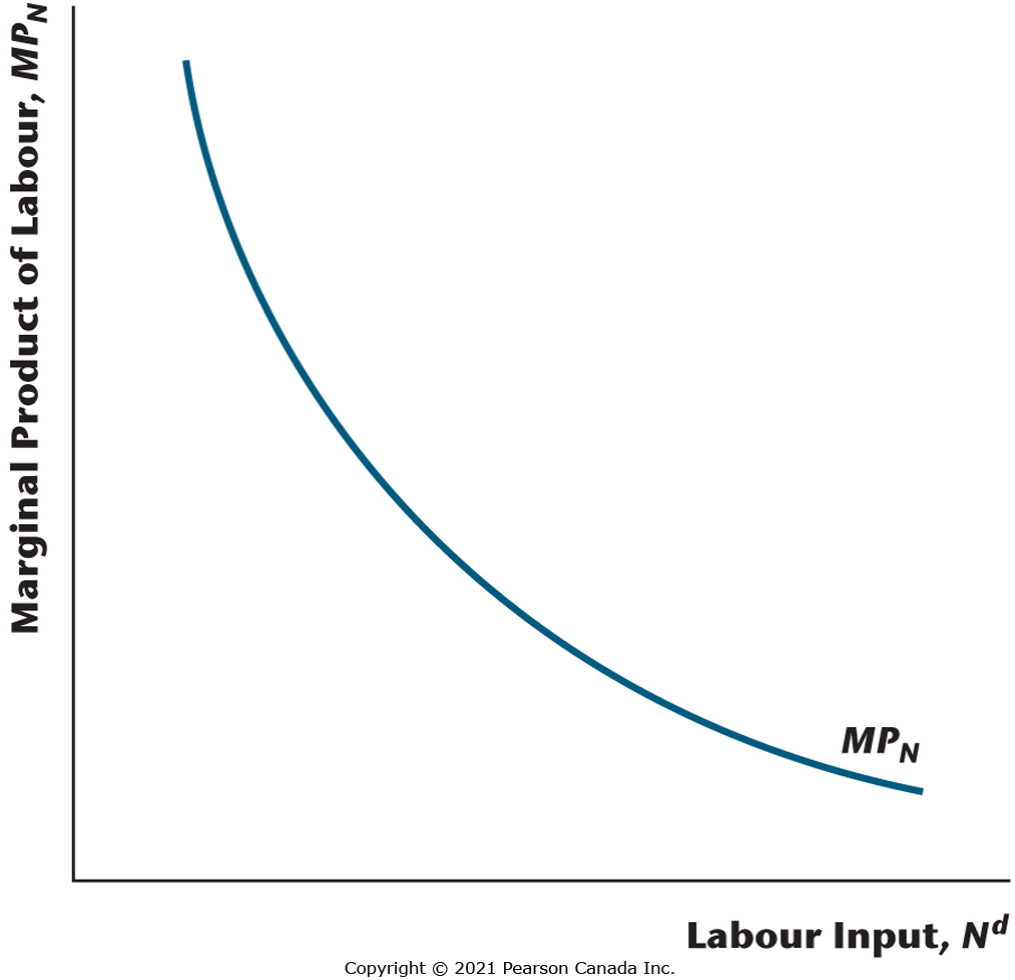
$$zF(xK, xN^d) = xzF(K, N^d) = xY. \tag{1}$$

  - e.g., let $x = 2$. output doubles if all factor inputs double.
- There are production functions exhibiting increasing returns to scale or decreasing returns to scale, which we do not consider.
- The CRS is an assumption, it is also factual and fits the aggregate economy.
- Given a CRS production function, the economy with a **representative firm** behaves in exactly the same way as if there were **many small firms** producing the same good with the same TFP z.

- Suppose, the economy has 3 firms, all producing the same good with the same z.
    - The first firm's inputs are K and N^d, output is Y.
    - The second firm's capital and labor inputs are $2K$ and $2N^d$.
    - The third firm: $3K$ and $3N^d$.
- Total output in the economy is then $6Y$ ($= Y + 2Y + 3Y$).

- Now assume that the economy has only a representative firm with capital $6K$ and $6N^d$, it produces the same good with the TFP z.
    - Output is $6Y$: $zF(6K, 6N^d) = 6zF(K, N^d) = 6Y$.
- Thus, with a CRS production function, we only need to study the optimal production by the representative firm.

- **Property 2.** Output increases if capital or labor input increases, or both increase.
- This says that $F()$ is increasing in K and N^d.
- We already saw this property, it implies that $MP_k > 0$ and $MP_N > 0$.
- **Property 3.** The marginal product of labor input decreases as the quantity of labor increases, i.e., MP_N becomes smaller as N^d increases.
    - But, the decline in MP_N never makes $MP_N < 0$.
- **Property 4.** The marginal product of capital input decreases as the capital stock becomes larger, i.e., MP_K becomes smaller as K increases.
    - But, the decline in MP_K never makes $MP_K < 0$.



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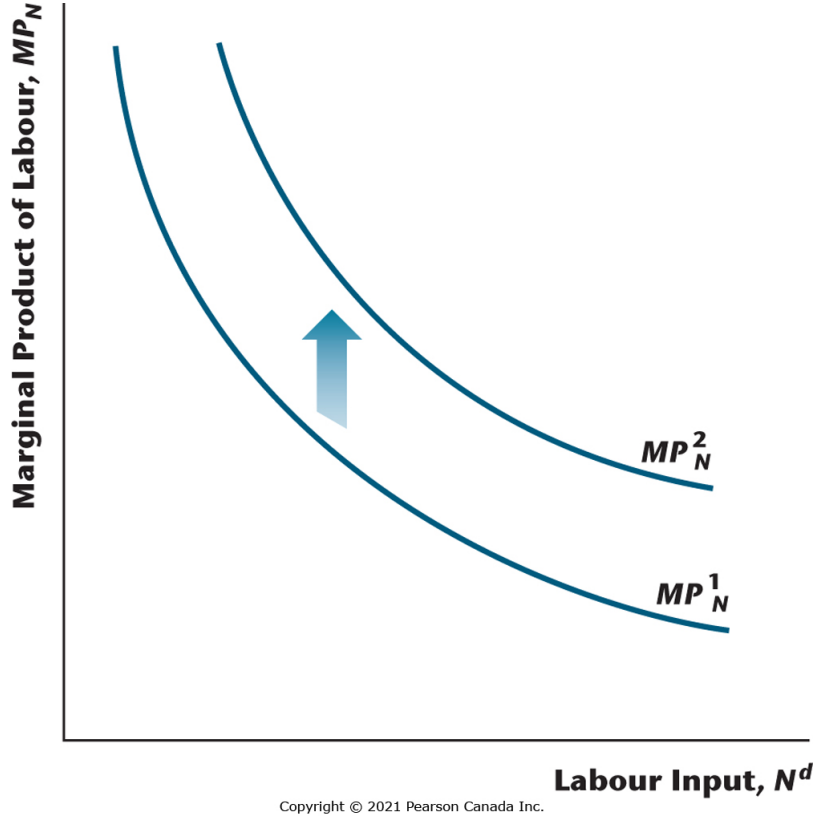
- **Property 5.** The marginal product of labor input increases as the quantity of capital stock increases.



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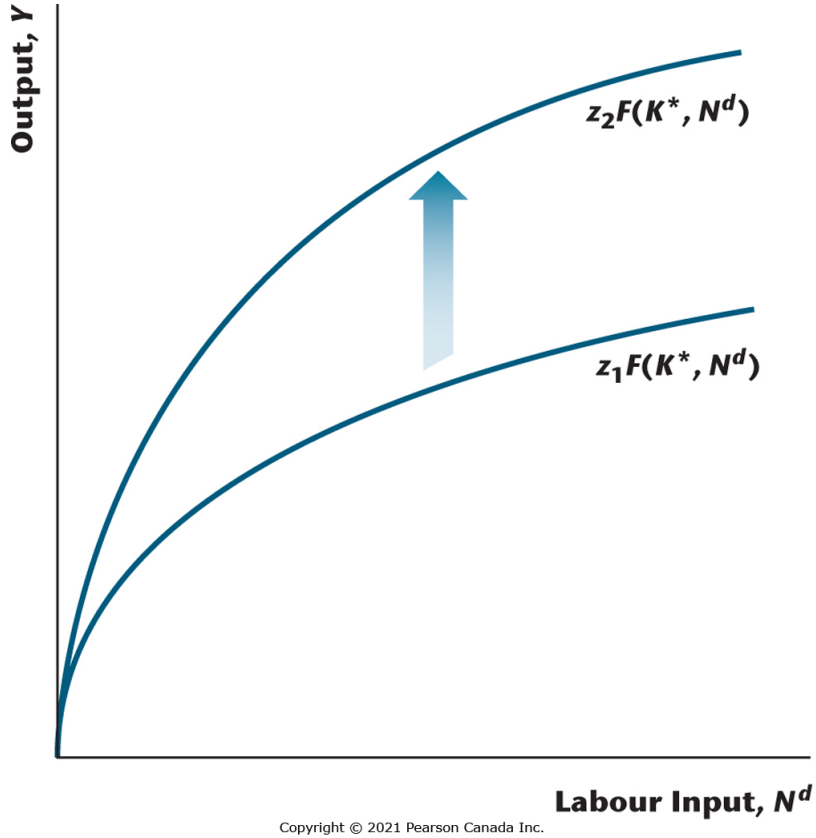
2-2. TOTAL FACTOR PRODUCTIVITY AND ITS EFFECT ON OUTPUT

- TFP summarizes factors that can change output while factor inputs remain unchanged.
- If we knew and knew how to measure these factors other than capital and labor, they would have been a factor of production explicitly.
- Some people call TFP a "measure of our ignorance" about production.
- One important factor causing z to increase is technological innovation.
- Management practice is another example that can improve TFP.
- Yet, researches find that government regulation is also a factor affecting TFP.

- If TFP becomes higher, then both output and marginal products increase



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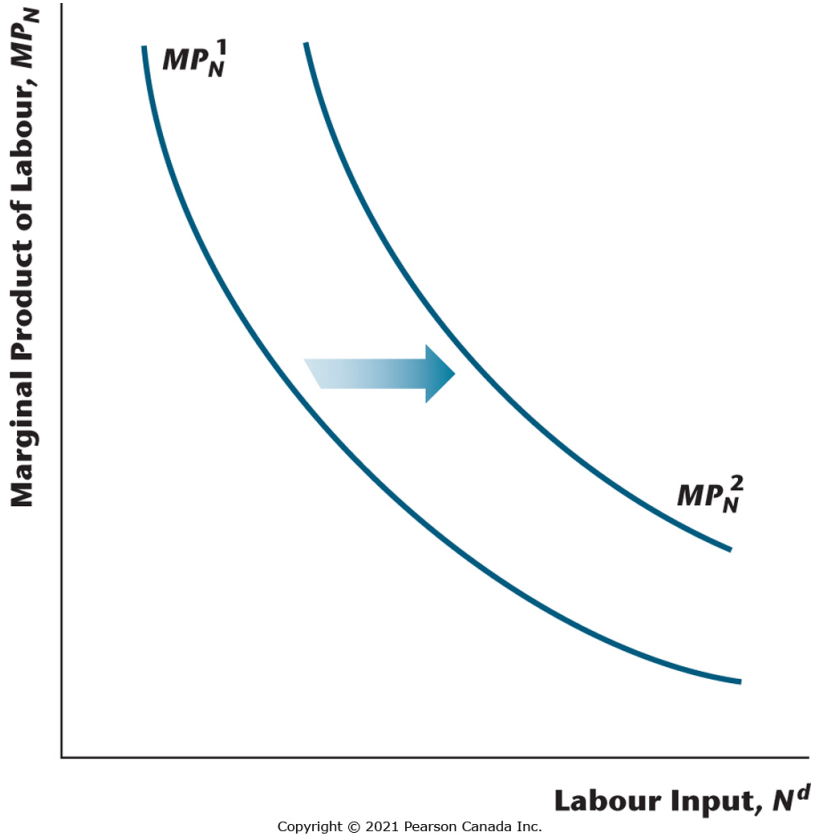
- In this graph, it should be: MP_N shifts **upward**, not **rightward**.



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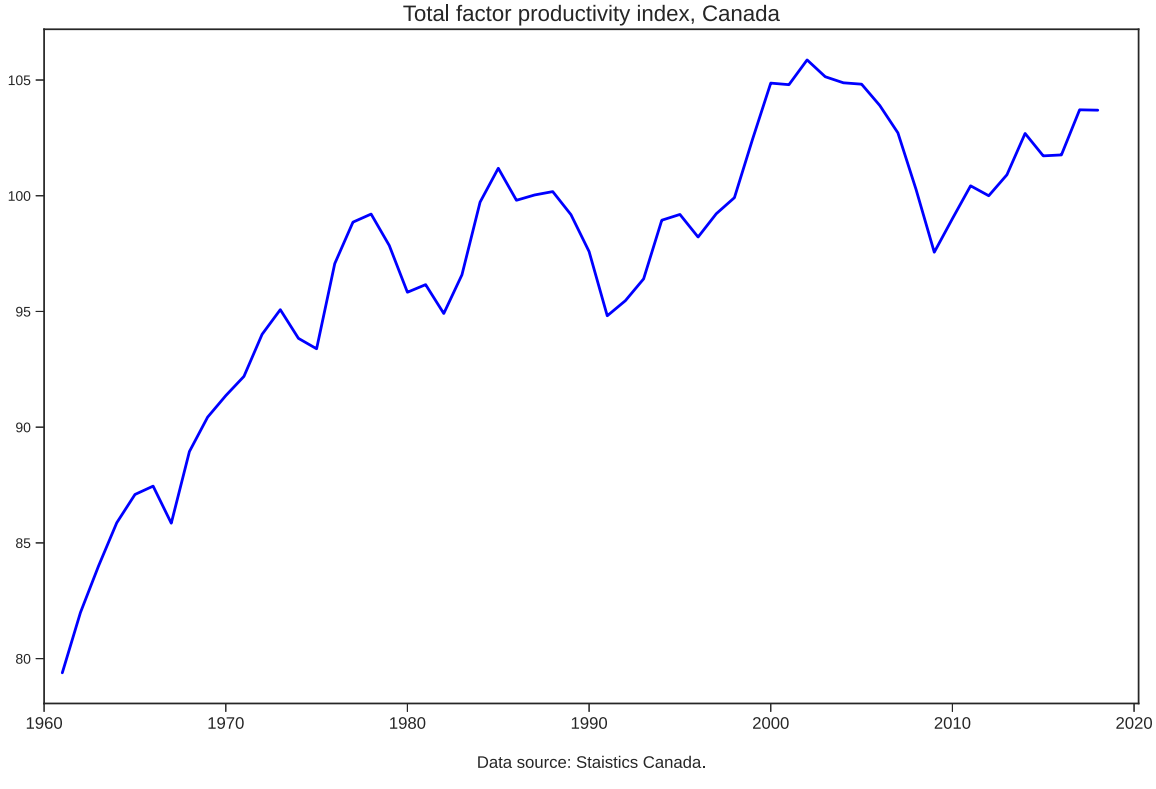
- TFP data in Canada.



2-3. PROFIT MAXIMIZATION

- The representative firm's optimization problem is to maximize profit, taking prices as given.
- For now, we fix the capital (assuming it is given).
- The nominal variable profit is $PY - WN^d$.
    - P is output price, Y is quantity of output.
    - W is the money wage.
- Re-write the nominal variable profit as $P\left(Y - \frac{W}{P}N^d\right)$.
- $\frac{W}{P} = w$ is the real wage (nominal wage relative to output price).
    - If all output is used for consumption C, then output price is the consumption price.

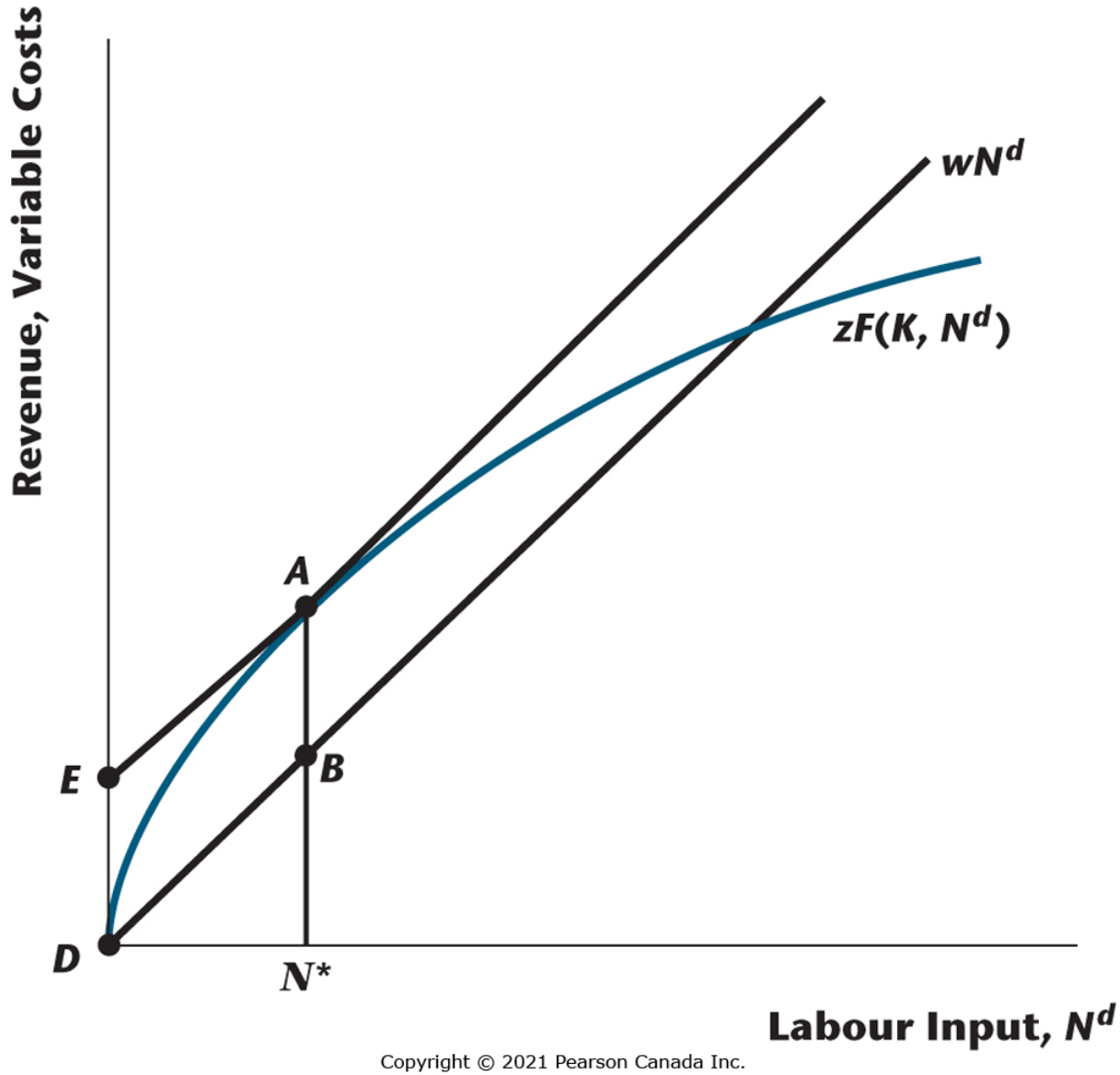
- Clearly, nominal variable profit is maximized if $\pi = Y - wN^d$ is maximized.

- **Thus, the representative firm's optimization problem is to choose the optimal labor input to maximize real value of variable profit π, taking the real wage as given**.

- Substituting production function for Y, real profit is $\pi = zF(K, N^d) - wN^d$.

    - π the real variable profit, since K is fixed and labor cost wN^d is a variable cost.
    - This often applies to a short-run horizon.
- Real variable profit is maximized when the difference (distance) between $zF(K, N^d)$ and wN^d is the largest.

- The optimal labor input can be characterized on a graph, see next page.



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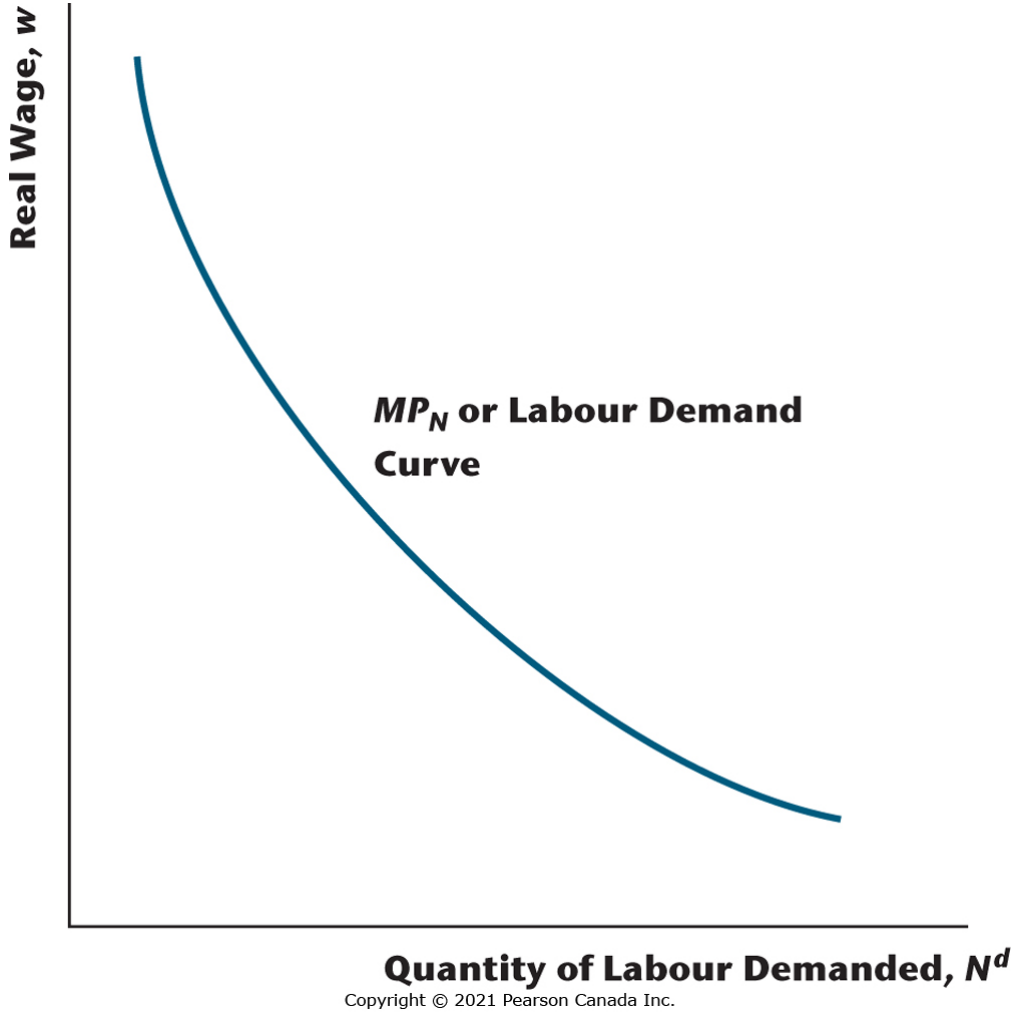
- At point A, the slope of production function is the marginal product of labor.
- The tangent line EA is parallel to the cost line DB which has a slope w.
- Thus, **real variable profit is maximized when the marginal product of labor is equal to the real wage rate.** That is,

$$MP_N = w. \tag{2}$$

- Intuition: $MP_N - w$ is the addition to profits when the firm adds one additional unit of labor input. The firm will keep adding more units of labor input, and stop when adding an additional unit of labor input no longer generate additional profits.
- When $N^d = N^*$, the firm adds no more labor, since adding another unit of labor creates a profit loss.
- When $MP_N = w$, we obtain the optimal choice of labor input.

THE LABOR DEMAND CURVE

- The labor demand curve describes the optimal choice of labor input given each level of real wage.
- Thus, the optimality condition of maximizing the real variable profit provides the labor demand curve.
- Mathematically, $MP_N = w$ is the labor demand curve.
- Question: what is the effect of an increase in z on the labor demand curve?



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